

1 / 5 | Ages 5-6 | 20 minutes

Strong paper bridges

What we are practising

Change a structure's shape and compare how it carries a load.

What you need

Two identical paper sheets / Two equal-height books / Large lightweight toy car

Use a large empty cardboard box as a light load.

Before you begin

Place books about a hand-width apart on the floor.

Build low on the floor; an adult handles heavy books and the bridge never supports a person.

Try it together

1. Lay a flat sheet across the books and gently place the toy in the middle.
2. Fold the second sheet into accordion pleats and span the same gap.
3. Test the same toy in the same position and compare bending.

Talk, draw or show your idea

Did changing the shape change the result?

Space to explore

2 / 5 | Ages 5-6 | 15 minutes

Which cloth drinks water?

What we are practising

Observe absorption while keeping starting conditions similar.

What you need

Cotton cloth / Smooth plastic lid / Tray / Teaspoon / Small cup of water

Compare paper towel with an unbreakable plate.

Before you begin

Put two dry surfaces on a tray; an adult measures the water.

An adult supervises water, keeps it from electrical items, wipes spills and empties the cup afterwards.

Try it together

1. Predict which surface will soak up water.
2. Put one teaspoon of water on each surface and wait a minute.
3. Look for spreading dampness or a separate puddle, then discuss a useful cleaning material.

Talk, draw or show your idea

Which material would you choose for a small spill?

Space to explore

3 / 5 | Ages 5-6 | 15 minutes

Leaf observer notebook

What we are practising

Describe similarities and differences based on observation.

What you need

Paper / Pencil / Two safe plants viewed with an adult

Use clear leaf photographs in a book.

Before you begin

Choose familiar non-irritating plants; look without picking or tasting.

An adult chooses plants and location; no tasting, picking unknown plants or touching sap. Wash hands after handling plants.

Try it together

1. Sketch one leaf's outline and visible lines.
2. Observe a second leaf and sketch it beside the first.
3. Circle one similarity and describe two differences such as width or edge shape.

Talk, draw or show your idea

Which details can we actually see rather than guess?

Space to explore

4 / 5 | Ages 5-6 | 15 minutes

Let the light through

What we are practising

Observe that different materials transmit different amounts of light.

What you need

Torch / Clear unbreakable lid / Baking paper / Cardboard / Wall

Use daylight at a window and look at objects through each material without looking at the sun.

Before you begin

An adult holds the torch aimed at the wall at a fixed distance.

An adult controls the torch; never aim at eyes or look at the sun. Use no glass or plastic bags.

Try it together

1. Hold the clear lid in the beam and observe the wall.
2. Replace it with baking paper and then cardboard in the same position.
3. Describe which lets a clear beam through, scatters it, or blocks it.

Talk, draw or show your idea

Which would make a better light-blocking cover?

Space to explore

5 / 5 | Ages 5-6 | 10 minutes

Feel a humming sound

What we are practising

Notice vibration associated with making a voice sound.

What you need

No materials; a paper sheet is optional for notes

Watch an adult demonstrate and describe what they feel instead of touching your own throat.

Before you begin

Sit comfortably; participation and touching are optional.

Touch only your own throat lightly; never press, hold breath or force sound. Stop with any discomfort.

Try it together

1. Rest fingertips lightly on your own throat without pressing.
2. Hum gently for a short moment and notice any vibration.
3. Stop and breathe normally; compare and repeat once if comfortable.

Talk, draw or show your idea

What changed when you stopped humming?

Space to explore

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

1. Strong paper bridges

A little hint

Keep the gap and load the same for both tests.

Answer / what to notice

A pleated sheet often bends less; record your actual result if both hold or both collapse.

Folds can make paper resist bending without adding more paper.

Make it easier

An adult folds while the child tests and observes.

Take it further

Try another fold shape with a fresh identical sheet.

Another day, away from the screen

Look for folded or ridged cardboard in packaging.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

2. Which cloth drinks water?

A little hint

Check whether the water stays separate or enters the material.

Answer / what to notice

Cotton usually absorbs water; smooth plastic keeps it on the surface. Coatings can change results.

Absorbent materials take water into small spaces within them.

Make it easier

Observe one drop-sized splash on each surface with adult help.

Take it further

Repeat with dry samples and two equal spoonfuls each.

Another day, away from the screen

Choose a suitable cloth to help clean the tray.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

3. Leaf observer notebook

A little hint

Look at the outline, colour and lines on each leaf.

Answer / what to notice

Responses depend on the leaves; accept observations tied to visible features, not guessed plant names.

A scientific description separates what we observe from what we assume.

Make it easier

Compare only broad and narrow outlines.

Take it further

Draw a third leaf and group it using a stated feature.

Another day, away from the screen

Revisit one plant later and record any visible change.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

4. Let the light through

A little hint

Compare the brightness on the wall rather than staring at the torch.

Answer / what to notice

Clear plastic transmits much light; baking paper scatters some; cardboard blocks most direct light.

The material affects how much light passes through and how it spreads.

Make it easier

Compare only clear plastic and cardboard.

Take it further

Predict what two layers of baking paper will do, then test.

Another day, away from the screen

Find a window, curtain and wall and discuss their different jobs.

For the grown-up

Use these prompts when useful. Open-ended activities can have many good answers. Let your child explain their thinking.

5. Feel a humming sound

A little hint

Focus on a gentle buzzing feeling, not loudness.

Answer / what to notice

You may feel vibration during humming and less during quiet breathing; it can be subtle.

Vocal folds vibrate to produce voiced sounds such as humming.

Make it easier

Listen while the adult describes the sensation.

Take it further

Compare a gentle hum with a softly spoken vowel.

Another day, away from the screen

Listen for voices and instruments in a familiar song.